



# Effect of semaglutide on kidney outcomes in the SELECT, FLOW, and SOUL trials: a prespecified pooled analysis

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## Summary

**Background** The GLP-1 receptor agonist semaglutide reduces clinically important kidney outcomes in people with type 2 diabetes and chronic kidney disease (CKD). We aimed to assess the pooled effects of semaglutide on kidney outcomes in prespecified analyses of participant-level data from the diverse populations of the SELECT, FLOW, and SOUL randomised placebo-controlled trials.

**Methods** Participants with CKD (FLOW) or atherosclerotic cardiovascular disease (SELECT and SOUL) were randomly assigned semaglutide (once-weekly subcutaneous 1.0 mg [FLOW], once-weekly subcutaneous 2.4 mg [SELECT], or once-daily oral 14 mg [SOUL]) or matching placebo, added to standard of care. The primary outcome in this pooled analysis was time to first occurrence of a kidney composite, defined as onset of persistent 50% or greater reduction in estimated glomerular filtration rate (eGFR), kidney failure (persistent eGFR <15 mL/min per 1.73 m<sup>2</sup>, or initiation of kidney replacement therapy), kidney-related death, or cardiovascular-related death. Safety was also assessed.

**Findings** The pooled participants from the trials (N=30787) had a mean follow-up of 39.5–47.5 months. Among participants assigned to semaglutide, 973 first events of the primary kidney composite occurred compared with 1134 first events for placebo (hazard ratio [HR] 0.84 [95% CI 0.77–0.91]). First events of a narrower secondary kidney composite (excluding cardiovascular-related death from the primary outcome) were also reduced with semaglutide versus placebo (347 and 416, respectively; 0.80 [0.69–0.92]). Safety outcomes were overall similar between groups, and in line with other GLP-1 receptor agonist trials. Serious adverse events were numerically lower with semaglutide than with placebo.

**Interpretation** Data pooled from three large phase 3 trials suggest that semaglutide reduces the risk of major kidney outcomes in a broad population with cardio-kidney-metabolic disease while having a favourable risk-benefit profile. In people with cardio-kidney-metabolic disease, with and without diabetes, semaglutide (oral or injected) prevents kidney-related and cardiovascular complications and induces adverse events in line with GLP-1 receptor agonist studies, regardless of baseline characteristics within the cardio-kidney-metabolic spectrum. This was a participant-level analysis conducted in a large database of three randomised controlled trials of similar design and examining the same treatment, but there were some differences in participants' baseline characteristics, and in the dose and route of administration of treatment. This pooled analysis adds evidence for the benefit of GLP-1 receptor agonists in general, and semaglutide in particular, in a broad population of people with cardio-kidney-metabolic disease, suggesting that the benefit of semaglutide might not be explained only by its glycaemic effects, weight-management effects, or both.

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## Introduction

Chronic kidney disease (CKD) is an important and growing cause of morbidity and mortality globally.<sup>1–3</sup> The increase in CKD prevalence is attributable to the increased prevalence of the contributing comorbidities of obesity, diabetes, hypertension, and cardiovascular diseases.<sup>4–6</sup> In addition, progress in the prevention of premature death from atherosclerotic cardiovascular disease (ASCVD) has increased the life expectancy of people living with obesity and type 2 diabetes; consequently, the prevalence of other age-related complications, including CKD, has risen.<sup>7</sup>

GLP-1 receptor agonists have been extensively investigated for the management of people with type 2 diabetes, obesity, or both. Results from cardiovascular outcomes trials have shown that several GLP-1 receptor agonists reduce the risk of major adverse cardiovascular events (MACE) and might have kidney-protective effects.<sup>8,9</sup> Published meta-analyses have summarised the beneficial effects of GLP-1 receptor agonists on cardiovascular and kidney outcomes.<sup>10,11</sup> However, the aggregate dataset included a minority of participants with CKD, and kidney effects were examined as secondary or post-hoc outcomes and were not

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## Research in context

### Evidence before this study

We searched PubMed using the terms “Glucagon-like peptide-1 receptor agonist (GLP-1RA)”, “Chronic kidney disease (CKD)”, “Atherosclerotic cardiovascular disease (ASCVD)”, “Type 2 diabetes”, “Obesity”, “Kidney outcomes”, and “Cardiovascular (CV) outcomes” for all articles published in English before May 1, 2025. Results from previously published meta-analyses have shown that, compared with placebo, GLP-1 receptor agonists substantially reduce the risk of clinically important cardiovascular and kidney outcomes in people with chronic kidney disease (CKD), atherosclerotic cardiovascular disease, or risk factors including type 2 diabetes and overweight or obesity. However, these meta-analyses were primarily based on cardiovascular outcome trials and were limited by lack of participant-level data, lack of uniformity of kidney outcomes (sometimes including changes in albuminuria), and absence of analyses according to important subgroups such as baseline estimated glomerular filtration rate (eGFR; eg, <60 mL/min per 1.73 m<sup>2</sup> vs ≥60 mL/min per 1.73 m<sup>2</sup>). In addition, with the exception of the FLOW trial, these trials included a relatively small number of participants with CKD.

### Added value of this study

Individual participant-level pooled analyses across the SELECT, FLOW, and SOUL placebo-controlled trials provide a robust assessment of the effect of semaglutide on kidney-related outcomes, similarly defined in the three trials, across disease states (people with and without diabetes and cardiovascular disease, CKD, or both). Analyses of participant-level data allow for assessment of time-to-event comparisons of identically defined kidney outcomes. The results indicate that semaglutide improves kidney-related outcomes, with benefits apparent with both subcutaneous and oral formulations. Consistent kidney benefits of semaglutide were observed across most participant subgroups, for example, those defined by baseline eGFR, urinary albumin-to-creatinine ratio, HbA<sub>1c</sub>, or BMI.

### Implications of all the available evidence

In this prespecified analysis, we found semaglutide was effective in decreasing CKD progression regardless of the administration route or indication (type 2 diabetes or obesity), adding to the evidence provided by FLOW in people with type 2 diabetes and CKD. These results could affect clinical decision making in the respective populations.

uniformly defined.<sup>10</sup> Subgroup analyses of potential clinical relevance, based on important characteristics such as baseline kidney function, were therefore not possible due to a lack of individual participant data.

FLOW (Evaluate renal function with semaglutide once weekly) is a large-scale clinical trial of kidney outcomes in people with CKD and type 2 diabetes following treatment with the GLP-1 receptor agonist semaglutide. Compared with placebo, semaglutide (once-weekly subcutaneous 1.0 mg) significantly reduced the risk of clinically important kidney outcomes, MACE, and all-cause mortality, but was not powered to assess effects on kidney failure alone or key subgroups.<sup>12</sup> Therefore, we conducted pooled analysis of the effects of semaglutide on kidney outcomes and safety outcomes in FLOW and a further two trials with a long-term mean follow-up beyond 3 years: SELECT (Semaglutide effects on cardiovascular outcomes in people with overweight or obesity)<sup>13</sup> and SOUL (Semaglutide cardiovascular outcomes trial)<sup>14</sup> cardiovascular outcomes trials. In these trials, populations were enriched for ASCVD prevalence and risk with or without type 2 diabetes. The effect of semaglutide, given at various doses and via different routes of administration, was assessed on adjudicated kidney outcomes that were similarly predefined in the FLOW, SELECT, and SOUL clinical trials.<sup>12–14</sup> Previous cardiovascular outcomes trials of semaglutide had a much shorter follow-up (≤2 years), kidney outcomes were analysed post hoc, not adjudicated,<sup>15</sup> and were primarily driven by changes in albuminuria and estimated glomerular filtration rate (eGFR) given the

short follow-up.<sup>16,17</sup> Given the considerable heterogeneity of the populations across the three trials, we present a summary of effects across participants with CKD and/or cardiovascular disease combined with type 2 diabetes, obesity, or both.

## Methods

### Study design

This study is a prespecified pooled analysis of the CKD endpoints of the SELECT, FLOW, and SOUL trials, the full details of the trial designs of which have been published previously;<sup>12–14</sup> individual trial designs are summarised in appendix 1 (p 10). Study and trial protocols and statistical analysis plans, including for this analysis, are in appendix 2. All three trials were of randomised, double-blind, parallel-group, placebo-controlled design, comparing semaglutide—in subcutaneous or oral form—with placebo added to standard of care. Participants provided written informed consent. All trials were event-driven, with trial closure being performed when the targeted number of primary endpoint events was reached or following the recommendation of the data monitoring committee.<sup>12–14</sup> In FLOW, the data monitoring committee recommended early completion of the trial for efficacy following a preplanned interim analysis.<sup>12</sup>

### Participants

This prespecified pooled analyses study comprised individual participant-level data from the full analysis set (FAS; defined as all randomly assigned participants and

See Online for appendix 1

See Online for appendix 2

grouped according to treatment assigned at randomisation) of the SELECT, FLOW, and SOUL trials combined, where the treatment groups were pooled to semaglutide or placebo. Participants were recruited between Oct 31, 2018, and May 3, 2021. According to the eligibility criteria of each trial, populations were enriched for CKD in FLOW and for ASCVD in SELECT and SOUL. FLOW participants were aged 18 years and older, with a diagnosis of type 2 diabetes ( $\text{HbA}_{1c} \leq 10.0\%$ ) and with high-risk CKD receiving a stable maximal labelled or tolerated dose of a renin-angiotensin system (RAS) inhibitor (angiotensin-converting enzyme inhibitor or angiotensin receptor blocker).<sup>12</sup> In SELECT, participants were aged 45 years and older, without a history of diabetes but with a BMI of at least 27 kg/m<sup>2</sup> and a history of cardiovascular disease (defined as previous myocardial infarction, previous stroke, or symptomatic peripheral arterial disease).<sup>13</sup> In SOUL, participants were aged 50 years and older, with a diagnosis of type 2 diabetes ( $\text{HbA}_{1c}$  6.5–10.0%) and established ASCVD, CKD, or both.<sup>14</sup> Sex, race, and ethnicity were self-reported.

## Procedures

For each trial, participants received semaglutide (once-weekly subcutaneous 1.0 mg [FLOW], once-weekly subcutaneous 2.4 mg [SELECT], once-daily oral 14 mg [SOUL]) or matching placebo in addition to standard of care in accordance with local or regional guidance.<sup>12–14</sup> In FLOW, a stable dose of an RAS inhibitor was mandated unless contraindicated or not tolerated.<sup>12</sup> Use of SGLT2 inhibitors and mineralocorticoid receptor antagonists (MRAs) was permitted, and randomisation was stratified according to baseline use of SGLT2 inhibitors in FLOW. Participants in the SELECT trial were not receiving SGLT2 inhibitors because these drugs were not indicated in people without diabetes at the time of the trial. Baseline use of SGLT2 inhibitors was reported in 16% of participants in FLOW and 27% in SOUL. Full details of participants' baseline medications can be found in the respective primary publications.<sup>12–14</sup>

All three trials were event driven; FLOW ended prematurely on the recommendation of the Independent Data Monitoring Committee. Data from the three trials were taken for pooled analysis on Mar 21, 2025, after the trials had ended.

## Outcomes

For these pooled analyses, the primary kidney outcome was time to first occurrence of a kidney composite, defined as persistent (two consecutive measurements  $\geq 4$  weeks apart) 50% or greater reduction in eGFR (per Chronic Kidney Disease Epidemiology Collaboration 2009 [CKD-EPI] formula), kidney failure (defined as persistent eGFR  $< 15$  mL/min per 1.73 m<sup>2</sup> or initiation of chronic kidney replacement therapy [dialysis or kidney transplantation]), or death from kidney-related or cardiovascular causes. This composite endpoint is identical to

the primary outcome in FLOW and the secondary kidney outcomes in SOUL and SELECT.<sup>12–14,18</sup> All components in the primary kidney outcome, except eGFR assessments, were adjudicated in a blinded fashion by an event adjudication committee.

Secondary kidney outcomes in this analysis were time to first occurrence of a narrower kidney composite on the basis of the criteria of the primary kidney outcome but excluding cardiovascular-related death; time to first occurrence of a kidney failure outcome, defined as onset of persistent eGFR of less than 15 mL/min per 1.73 m<sup>2</sup>, initiation of chronic kidney replacement therapy (dialysis or kidney transplantation), or both; time to first occurrence of the individual components of the primary kidney composite; time to first occurrence of the kidney composite outcomes defined according to different eGFR reductions ( $\geq 30\%$ ,  $\geq 40\%$ , and  $\geq 57\%$ ); and annual rate of change in eGFR.

The annual rate of change in eGFR (total eGFR slope) was assessed according to baseline eGFR ( $< 60$  mL/min per 1.73 m<sup>2</sup> and  $\geq 60$  mL/min per 1.73 m<sup>2</sup>), urinary albumin-to-creatinine ratio (UACR;  $< 30$  mg/g,  $\geq 30$  mg/g to  $\leq 300$  mg/g, and  $> 300$  mg/g in FLOW and SELECT only, as UACR was not measured in SOUL), Kidney Disease Improving Global Outcomes (KDIGO) risk category (low, moderate, high, very high; FLOW and SELECT only), and baseline use of SGLT2 inhibitors. Similarly, changes in eGFR from baseline to week 104 were assessed according to baseline CKD severity (eGFR, UACR, and KDIGO risk category) and baseline SGLT2 inhibitor use.

Participant subgroups analysed for the primary and narrower secondary kidney composites were sex, age ( $< 65$  years vs  $\geq 65$  years), BMI ( $< 30$  kg/m<sup>2</sup> vs  $\geq 30$  kg/m<sup>2</sup>), high-sensitivity C-reactive protein (hsCRP) concentration ( $< 2$  mg/L,  $\geq 2$  mg/L to  $< 10$  mg/L, and  $\geq 10$  mg/L),  $\text{HbA}_{1c}$  level ( $< 7\%$  vs  $\geq 7\%$ ), KDIGO risk category (low, moderate, high, or very high; FLOW and SELECT only), type 2 diabetes (yes or no), UACR ( $< 30$  mg/g,  $\geq 30$  mg/g to  $\leq 300$  mg/g, and  $> 300$  mg/g; FLOW and SELECT only), eGFR ( $< 60$  mL/min per 1.73 m<sup>2</sup> vs  $\geq 60$  mL/min per 1.73 m<sup>2</sup> and  $< 45$  mL/min per 1.73 m<sup>2</sup>,  $\geq 45$  mL/min per 1.73 m<sup>2</sup> to  $< 60$  mL/min per 1.73 m<sup>2</sup>, and  $\geq 60$  mL/min per 1.73 m<sup>2</sup>), and baseline use of SGLT2 inhibitors or MRAs.

Safety outcomes, including serious adverse events (SAEs) and adverse events leading to discontinuation, were summarised for the full analysis set and according to baseline eGFR ( $< 60$  mL/min per 1.73 m<sup>2</sup> and  $\geq 60$  mL/min per 1.73 m<sup>2</sup>). Grading of adverse events was as described in the individual trial protocols: the trial investigator was responsible for detecting, documenting, recording and following up on SAEs, adverse events requiring event adjudication or additional data collection (irrespective of seriousness), adverse events leading to discontinuation of trial product (irrespective of seriousness), adverse events of COVID-19 (irrespective of seriousness), pregnancies, and technical complaints.

### Data analysis

These pooled analyses were prespecified in a specific statistical pooled analysis plan ahead of database lock for the individual trials, and additional post-hoc analyses were also done. The full analysis set comprised participant-level data from the SELECT, FLOW, and SOUL trials combined. No alpha-spending was pre-specified for the pooled analyses; accordingly, these

analyses should be viewed as exploratory. The corresponding CIs are nominal and not intended as formal hypothesis tests. In each trial, an interim analysis was planned for superiority when two-thirds of the primary events had accrued. Of the three trials, only the FLOW trial was stopped at this interim analysis and no correction for potential overestimation of effect from this trial was performed. Detailed descriptions of the alpha-spending method used in each trial are provided in the respective primary publications.<sup>12–14</sup>

Trial biases were considered to be minor but could potentially include partial unblinding due to adverse events, differential dropout rates and background therapy changes.

For time-to-event outcomes, hazard ratios (HRs) for comparing the effects of semaglutide versus placebo were estimated from a stratified Cox proportional-hazards model with pooled treatment group (semaglutide or placebo) as a fixed factor, stratified by trial, together with the two-sided 95% CI. The exact ties method was used because it accounts for all possible orderings of tied events, which is important for kidney endpoints, including eGFR measurements, as these are mainly collected within specific visit windows during the trial. Cumulative incidence plots for the kidney composites used the Aalen–Johansen estimator. For the individual kidney outcomes, competing risks were modelled for non-cardiovascular-related and non-kidney-related death (for the primary kidney composite outcome), non-kidney-related death (for the narrower secondary kidney composite outcome), all-cause death (for the kidney failure and initiation of chronic kidney replacement therapy outcomes), and all-cause death and initiation of kidney replacement therapy (for the  $\geq 50\%$  reduction in eGFR and persistent eGFR  $< 15$  mL/min per  $1.73$  m<sup>2</sup> outcomes). For all relevant analyses, we calculated eGFR using the CKD-EPI 2009 formula, based on creatinine measurements.<sup>19</sup> In FLOW, eGFR measurements based on cystatin C were also collected.

The annual rate (slope) of change in eGFR was based on a linear random effects regression model using restricted maximum likelihood estimation and degrees of freedom using the containment method (which is generally conservative) with pooled treatment, trial, time (as a continuous variable), and treatment-by-time interaction as fixed effects, with participant effect as a random intercept and time as a random slope using observed values. An unstructured covariance matrix was used for assessing subgroup effects; subgroup, trial, treatment-by-subgroup interaction, and treatment-by-time-by-subgroup interaction were included as fixed effects. The random intercept and slope were assumed to be bivariate normally distributed, with mean zero and an unstructured covariance matrix. The independent error term was assumed to be univariate normally distributed with mean zero. The model was fitted to observed scheduled data, excluding eGFR data from confirmatory

|                                           | SELECT<br>(n=17 604) | SOUL<br>(n=9650)    | FLOW<br>(n=3533)        | Total<br>(N=30 787) |
|-------------------------------------------|----------------------|---------------------|-------------------------|---------------------|
| Sex                                       |                      |                     |                         |                     |
| Female                                    | 4872 (27.7%)         | 2790 (28.9%)        | 1069 (30.3%)            | 8731 (28.4%)        |
| Male                                      | 12732 (72.3%)        | 6860 (71.1%)        | 2464 (69.7%)            | 22 056 (71.6%)      |
| Age (years), median (IQR)                 | 61.0<br>(55.0–68.0)  | 66.0<br>(61.0–72.0) | 68.0<br>(61.0–73.0)     | 64.0<br>(57.0–70.0) |
| Age category                              |                      |                     |                         |                     |
| <65 years                                 | 10 876 (61.8%)       | 4034 (41.8%)        | 1285 (36.4%)            | 16 195 (52.6%)      |
| $\geq 65$ to <75 years                    | 5362 (30.5%)         | 4249 (44.0%)        | 1589 (45.0%)            | 11 200 (36.4%)      |
| $\geq 75$ years                           | 1366 (7.8%)          | 1367 (14.2%)        | 659 (18.7%)             | 3392 (11.0%)        |
| Ethnicity                                 |                      |                     |                         |                     |
| Hispanic or Latino                        | 1822 (10.3%)         | 1380 (14.3%)        | 556 (15.7%)             | 3758 (12.2%)        |
| Not Hispanic or Latino                    | 15 611 (88.7%)       | 8178 (84.7%)        | 2832 (80.2%)            | 26 621 (86.5%)      |
| Not reported                              | 171 (1.0%)           | 92 (1.0%)           | 145 (4.1%)              | 408 (1.3%)          |
| Race                                      |                      |                     |                         |                     |
| American Indian or Alaska Native          | 44 (0.2%)            | 19 (0.2%)           | 25 (0.7%)               | 88 (0.3%)           |
| Asian                                     | 1447 (8.2%)          | 2255 (23.4%)        | 846 (23.9%)             | 4548 (14.8%)        |
| Black or African American                 | 671 (3.8%)           | 252 (2.6%)          | 160 (4.5%)              | 1083 (3.5%)         |
| Native Hawaiian or other Pacific Islander | 8 (<0.1%)            | 9 (0.1%)            | 5 (0.1%)                | 22 (0.1%)           |
| White                                     | 14 791 (84.0%)       | 6648 (68.9%)        | 2323 (65.8%)            | 23 762 (77.2%)      |
| Other                                     | 474 (2.7%)           | 377 (3.9%)          | 95 (2.7%)               | 946 (3.1%)          |
| Not reported                              | 169 (1.0%)           | 90 (0.9%)           | 79 (2.2%)               | 338 (1.1%)          |
| Geographical region                       |                      |                     |                         |                     |
| Asia                                      | 2201 (12.5%)         | 2478 (25.7%)        | 912 (25.8%)             | 5591 (18.2%)        |
| Europe                                    | 6692 (38.0%)         | 2887 (29.9%)        | 963 (27.3%)             | 10 542 (34.2%)      |
| North America                             | 4401 (25.0%)         | 1912 (19.8%)        | 865 (24.5%)             | 7178 (23.3%)        |
| Other                                     | 4310 (24.5%)         | 2373 (24.6%)        | 793 (22.4%)             | 7476 (24.3%)        |
| Smoking status                            |                      |                     |                         |                     |
| Current smoker                            | 2950 (16.8%)         | 1129 (11.7%)        | 429 (12.1%)             | 4508 (14.6%)        |
| Previous smoker                           | 8532 (48.5%)         | 4290 (44.5%)        | 1357 (38.4%)            | 14 179 (46.1%)      |
| Never smoked                              | 6120 (34.8%)         | 4230 (43.8%)        | 1747 (49.4%)            | 12 097 (39.3%)      |
| Body weight (kg)                          | 96.7 (0.13)          | 87.9 (0.20)         | 89.6 (0.35)             | 93.1 (0.11)         |
| BMI (kg/m <sup>2</sup> )                  | 33.3 (0.04)          | 31.1 (0.06)         | 32.0 (0.11)             | 32.5 (0.03)         |
| eGFR (mL/min per $1.73$ m <sup>2</sup> )  | 82.5 (0.13)          | 73.8 (0.23)         | 47.0 (0.26)             | 75.7 (0.13)         |
| UACR (mg/g), median (IQR)                 | 7.4<br>(4.5–15.4)    | ..                  | 567.6<br>(230.7–1296.7) | 9.4<br>(4.9–40.4)   |
| HbA <sub>1c</sub> (%)                     | 5.8% (0.00)          | 8.0% (0.01)         | 7.8% (0.02)             | 6.7% (0.01)         |
| HbA <sub>1c</sub> (mmol/mol)              | 39.7 (0.03)          | 63.5 (0.13)         | 61.5 (0.24)             | 49.6 (0.08)         |
| hsCRP (mg/L)                              | 3.9 (0.06)           | 4.2 (0.09)          | 5.5 (0.18)              | 4.2 (0.05)          |
| Total cholesterol (mmol/L)                | 4.2 (0.00)           | 4.0 (0.01)          | 4.4 (0.02)              | 4.2 (0.01)          |
| HDL cholesterol (mmol/L)                  | 1.2 (0.00)           | 1.1 (0.00)          | 1.1 (0.01)              | 1.1 (0.00)          |
| LDL cholesterol (mmol/L)                  | 2.2 (0.01)           | 2.0 (0.01)          | 2.3 (0.02)              | 2.2 (0.01)          |
| Triglycerides (mmol/L)                    | 1.8 (0.01)           | 2.1 (0.02)          | 2.4 (0.03)              | 1.9 (0.01)          |

(Table 1 continues on next page)

testing, which follows the approach taken in the FLOW and SOUL trials.<sup>12,14</sup> The estimated treatment difference (ETD) was the regression coefficient for the treatment-by-time interaction term, which measured the slope difference between semaglutide and placebo.

Post-hoc analyses of eGFR change from baseline to week 104 by subgroups was assessed using analysis of covariance, with pooled treatment, trial, subgroup, and treatment-by-subgroup interaction as fixed effects, and baseline values as covariates. Analysis of change in the full analysis set was done using the same model but without the treatment-by-subgroup interaction. UACR measurements were log-transformed before the analysis and the estimate and CIs were then back-transformed to obtain the estimated treatment ratio (ETR). Missing values at week 104 were accounted for with multiple imputation under a missing-at-random assumption, with the use of a common prespecified approach applied consistently across all three individual trials.<sup>12–14</sup>

As a planned supplementary analysis, the changes in eGFR and UACR from baseline to week 104 were analysed using a mixed model for repeated measures (MMRM). The model used data points from the in-trial observation period restricted to the first 104 weeks. The MMRM was fitted with treatment and trial as fixed factors and baseline value, BMI at baseline, and HbA<sub>1c</sub> at baseline as covariates, all nested within visit. Analysis of change in the full analysis set was done with the same model but without the treatment-by-subgroup interaction. An unstructured covariance matrix for measurements within the same participant was used. Degrees of freedom were calculated using the Kenward–Roger method.

Sensitivity analyses for the kidney composite outcomes were done post hoc to assess the effect of potential heterogeneity across trials. Trial-specific treatment effect estimates were derived separately and combined using both fixed-effect and random-effects meta-analytic models. Heterogeneity across trials was evaluated using the I<sup>2</sup> measure in the fixed-effect model. Safety data were summarised descriptively. No adjustments for multiplicity were made. A significance level of 5% was applied for the interaction p-values. All statistical analyses were performed with SAS software (version 9.4 TS1M5).

### Role of the funding source

Novo Nordisk provided funding for the statistical analysis and for the editorial support (medical writing assistance).

### Results

30787 participants were included (SELECT n=17604; FLOW n=3533; SOUL n=9650).<sup>12–14</sup> The mean duration of follow-up was 39.8 months (SD 9.4) in SELECT, 39.5 months (10.7) in FLOW, and 47.5 months (10.9) in SOUL. Inclusion criteria were different across trials; all participants had type 2 diabetes in FLOW and SOUL, and SELECT included participants with overweight or

|                                    | SELECT<br>(n=17 604) | SOUL<br>(n=9650)   | FLOW<br>(n=3533)    | Total<br>(N=30 787) |
|------------------------------------|----------------------|--------------------|---------------------|---------------------|
| (Continued from previous page)     |                      |                    |                     |                     |
| Duration of diabetes, median (IQR) | ..                   | 14.7<br>(9.0–20.8) | 16.3<br>(10.7–22.6) | 15.1<br>(9.6–21.0)  |
| Comorbid conditions                |                      |                    |                     |                     |
| Atrial fibrillation                | 1554 (8.8%)          | 662 (6.9%)         | 267 (7.6%)          | 2483 (8.1%)         |
| Hypertension                       | 14 392 (81.8%)       | 8759 (90.8%)       | 3460 (97.9%)        | 26 611 (86.4%)      |
| Coronary heart disease             | 14 452 (82.1%)       | 6821 (70.7%)       | 1331 (37.7%)        | 22 604 (73.4%)      |
| Coronary revascularisation         | 11 849 (67.3%)       | 5198 (53.9%)       | 691 (19.6%)         | 17 738 (57.6%)      |
| PCI                                | 10 337 (58.7%)       | 3755 (38.9%)       | 429 (12.1%)         | 14 521 (47.2%)      |
| CABG                               | 2057 (11.7%)         | 1730 (17.9%)       | 304 (8.6%)          | 4091 (13.3%)        |
| Heart failure                      | 4286 (24.4%)         | 2229 (23.1%)       | 678 (19.2%)         | 7193 (23.4%)        |
| Previous myocardial infarction     | 13 438 (76.3%)       | 3861 (40.0%)       | 514 (14.5%)         | 17 813 (57.9%)      |
| Previous stroke                    | 4110 (23.3%)         | 1488 (15.4%)       | 367 (10.4%)         | 5965 (19.4%)        |
| Concurrent medications             |                      |                    |                     |                     |
| β-blocker                          | 12 357 (70.2%)       | 6201 (64.3%)       | 1843 (52.2%)        | 20 401 (66.3%)      |
| SGLT2 inhibitor                    | 0 (0%)               | 2596 (26.9%)       | 550 (15.6%)         | 3146 (10.2%)        |
| Diuretic                           | 5900 (33.5%)         | 4064 (42.1%)       | 1780 (50.4%)        | 11 744 (38.1%)      |
| Loop diuretic                      | 2209 (12.5%)         | 1548 (16.0%)       | 890 (25.2%)         | 4647 (15.1%)        |
| MRA                                | 1820 (10.3%)         | 892 (9.2%)         | 257 (7.3%)          | 2969 (9.6%)         |
| Thiazides                          | 2027 (11.5%)         | 1575 (16.3%)       | 613 (17.4%)         | 4215 (13.7%)        |
| ACE inhibitor or ARB               | 13 054 (74.2%)       | 7618 (78.9%)       | 3364 (95.2%)        | 24 036 (78.1%)      |
| ARNI                               | 265 (1.5%)           | 35 (0.4%)          | 3 (0.1%)            | 303 (1.0%)          |

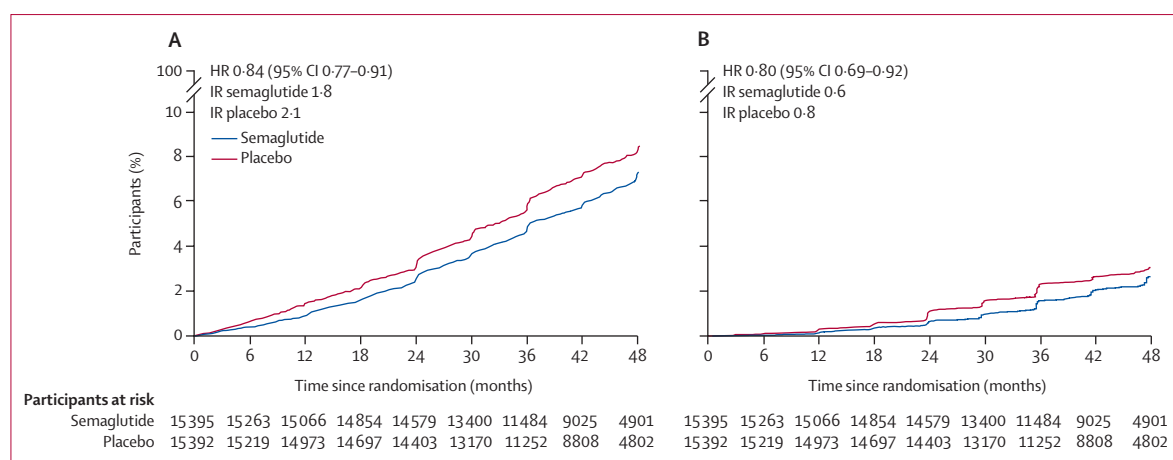
Data are n (%) or mean (SE) unless otherwise stated. UACR data were not collected in the SOUL trial. Type 2 diabetes was an exclusion criterion in SELECT. ACE=angiotensin-converting enzyme. ARB=angiotensin receptor blocker. ARNI=angiotensin receptor/neprilysin inhibitor. BMI=body mass index. CABG=coronary artery bypass graft. eGFR=estimated glomerular filtration rate. HDL=high-density lipoprotein. hsCRP=high-sensitivity C-reactive protein. IQR=interquartile range. LDL=low-density lipoprotein. MRA=mineralocorticoid receptor antagonist. ..=not applicable. PCI=percutaneous coronary intervention. SGLT2=sodium-glucose cotransporter-2. UACR=urinary albumin-to-creatinine ratio.

**Table 1: Baseline characteristics of the SELECT, SOUL, and FLOW trials in the full analysis set**

obesity but without diabetes. These differences are reflected in the participant baseline characteristics and use of concomitant medications shown in table 1. For example, participants in SELECT had higher bodyweight and lower HbA<sub>1c</sub> compared with participants in SOUL and FLOW, whereas participants in FLOW had the lowest eGFR and highest UACR of the three trials (table 1).

Participant disposition for each trial was reported in the primary publications (appendix 1 pp 11–14).<sup>12–14</sup> Across the three trials, at least 96.9% of participants completed the trial, and vital status was known for more than 97% of participants. Censoring before end of study for the primary kidney outcome was infrequent and broadly balanced between treatment groups, and visit-level missingness for eGFR was low overall (appendix 1 pp 2, 15–18).

In the pooled semaglutide group, there were fewer occurrences of the primary kidney composite outcome (973 first events, incidence rate [IR] 1.8 per 100 years of observation) compared with placebo (1134 first events, IR 2.1 per 100 years of observation; HR 0.84 [95% CI 0.77–0.91]; figure 1A, table 2). Individually, the SELECT,



**Figure 1: Cumulative incidence of kidney events over 48 months**

Cumulative incidence plots showing time from randomisation to occurrence of first primary kidney composite event (A) and narrower secondary kidney composite event (B). Pooled data from the SELECT, FLOW, and SOUL trials, in the full analysis set. The primary kidney composite included onset of persistent  $\geq 50\%$  reduction in eGFR, kidney failure (persistent eGFR  $< 15$  mL/min per  $1.73$  m $^2$  or initiation of kidney replacement therapy), kidney-related death, or cardiovascular-related death. Narrower secondary kidney composite included the above criteria, excluding cardiovascular-related death. eGFR=estimated glomerular filtration rate. HR=hazard ratio. IR=incidence rate (number of events per 100 years of observation).

|                                                                                             | Semaglutide<br>(n=15395) | Placebo<br>(n=15392) | Hazard ratio<br>(95% CI) |
|---------------------------------------------------------------------------------------------|--------------------------|----------------------|--------------------------|
| <b>Primary kidney outcome</b>                                                               |                          |                      |                          |
| Primary kidney composite                                                                    | 973 (6.3% [1.8])         | 1134 (7.4% [2.1])    | 0.84 (0.77–0.91)         |
| <b>Secondary kidney outcomes</b>                                                            |                          |                      |                          |
| Narrower secondary kidney composite<br>(excluding cardiovascular-related death)             | 347 (2.3% [0.6])         | 416 (2.7% [0.8])     | 0.80 (0.69–0.92)         |
| Kidney failure                                                                              | 206 (1.3% [0.4])         | 241 (1.6% [0.4])     | 0.83 (0.69–1.00)         |
| Persistent $\geq 50\%$ reduction in eGFR                                                    | 248 (1.6% [0.5])         | 320 (2.1% [0.6])     | 0.74 (0.63–0.88)         |
| Persistent eGFR $< 15$ mL/min per $1.73$ m $^2$                                             | 120 (0.8% [0.2])         | 147 (1.0% [0.3])     | 0.79 (0.62–1.00)         |
| Initiation of chronic kidney replacement<br>therapy (dialysis or kidney<br>transplantation) | 131 (0.9% [0.2])         | 154 (1.0% [0.3])     | 0.83 (0.65–1.04)         |
| Kidney-related death                                                                        | 6 (0.0% [0.0])           | 12 (0.1% [0.0])      | 0.49 (0.17–1.26)         |
| Cardiovascular death                                                                        | 647 (4.2% [1.2])         | 751 (4.9% [1.4])     | 0.85 (0.77–0.95)         |
| Persistent eGFR reduction of:                                                               |                          |                      |                          |
| $\geq 30\%$                                                                                 | 869 (5.6% [1.6])         | 1139 (7.4% [2.1])    | 0.72 (0.66–0.79)         |
| $\geq 40\%$                                                                                 | 455 (3.0% [0.8])         | 634 (4.1% [1.2])     | 0.68 (0.60–0.77)         |
| $\geq 57\%$                                                                                 | 144 (0.9% [0.3])         | 191 (1.2% [0.4])     | 0.73 (0.58–0.90)         |
| Primary kidney composite defined with<br>$\geq 40\%$ reduction in eGFR                      | 1153 (7.5% [2.1])        | 1394 (9.1% [2.6])    | 0.80 (0.74–0.86)         |
| Narrower secondary kidney composite defined with persistent:                                |                          |                      |                          |
| $\geq 30\%$ reduction in eGFR                                                               | 934 (6.1% [1.7])         | 1188 (7.7% [2.2])    | 0.74 (0.68–0.81)         |
| $\geq 40\%$ reduction in eGFR                                                               | 538 (3.5% [1.0])         | 693 (4.5% [1.3])     | 0.74 (0.66–0.83)         |
| $\geq 57\%$ reduction in eGFR                                                               | 269 (1.7% [0.5])         | 323 (2.1% [0.6])     | 0.81 (0.68–0.95)         |

Data are n (% [IR]), unless otherwise stated, and are from the in-trial period. Two-sided p-values not corrected for multiple comparisons. Primary kidney composite included onset of persistent  $\geq 50\%$  reduction in eGFR, kidney failure (persistent eGFR  $< 15$  mL/min per  $1.73$  m $^2$  or initiation of kidney replacement therapy), kidney-related death, or cardiovascular-related death. Narrower secondary kidney composite included the above criteria, excluding cardiovascular-related death. Kidney failure outcome included onset of persistent eGFR  $< 15$  mL/min per  $1.73$  m $^2$  and/or initiation of chronic kidney replacement therapy (dialysis or kidney transplantation). eGFR=estimated glomerular filtration rate. IR=incidence rate (number of events by 100 years of observation).

**Table 2: Primary and secondary time-to-first event kidney outcomes with pooled data from the SELECT, FLOW, and SOUL trials for semaglutide compared with placebo in the full analysis set**

FLOW, and SOUL trials contributed 528 events (semaglutide vs placebo: IR 0.8 vs 1.0), 741 events (5.6 vs 7.1), and 838 events (2.1 vs 2.3) to the primary kidney outcome, respectively.

Similarly, in participants treated with semaglutide, there were fewer first occurrences of the narrower secondary kidney composite (excluding cardiovascular-related death) compared with placebo (HR 0.80 [95% CI 0.69–0.92]; figure 1B; table 2). In the semaglutide group, 347 first events were reported compared with 416 in the placebo group. Corresponding IRs were 0.6 and 0.8 per 100 years of observation for semaglutide and placebo, respectively. Individually, the SELECT trial contributed 44 events (semaglutide vs placebo: IR 0.1 vs 0.1), FLOW contributed 478 events (IR 3.7 vs 4.5), and SOUL contributed 241 events (IR 0.6 vs 0.7) to the narrower kidney composite outcome.

For time to first occurrence of the kidney failure outcome, semaglutide treatment was associated with fewer first events than placebo (206 vs 241; HR 0.83 [95% CI 0.69–1.00]; figure 2; table 2). Corresponding IRs were 0.38 and 0.45 per 100 years of observation for semaglutide and placebo, respectively. HRs for time to first occurrence of the individual components of the primary kidney composite and for the first occurrence of the kidney composite outcomes according to varying reductions in eGFR ( $\geq 30\%$ ,  $\geq 40\%$ , and  $\geq 57\%$ ) are shown in table 2. Secondary kidney outcomes for the individual SELECT, FLOW, and SOUL trials are shown in appendix 1 (pp 3–5).

The annual decline in eGFR (total eGFR slope) favoured semaglutide ( $-1.26$  mL/min per  $1.73$  m $^2$  per year) versus placebo ( $-1.74$  mL/min per  $1.73$  m $^2$  per year), with an ETD of 0.48 mL/min per  $1.73$  m $^2$  per year (95% CI 0.40–0.56). In addition, the mean change in eGFR from

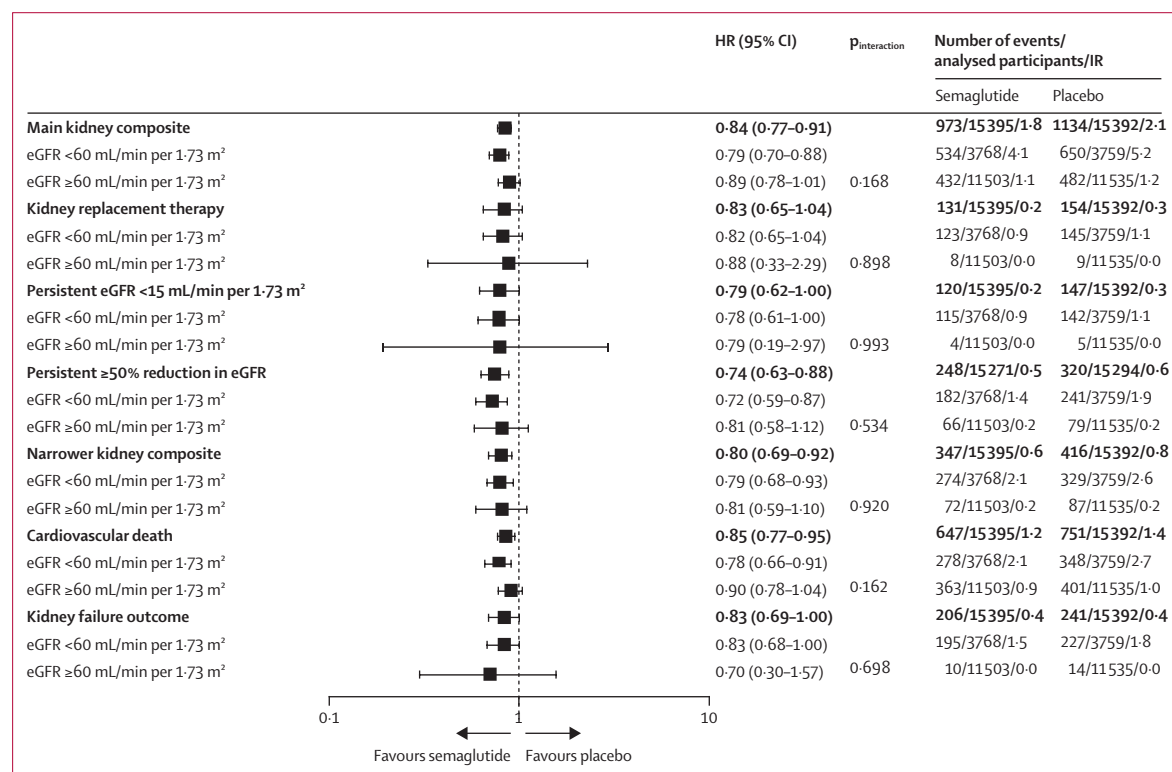
baseline to week 104 was  $-1.85$  mL/min per  $1.73$  m<sup>2</sup> for participants who received semaglutide and  $-2.93$  mL/min per  $1.73$  m<sup>2</sup> for those who received placebo. From baseline to week 104, the ETD between semaglutide and placebo was  $1.08$  mL/min per  $1.73$  m<sup>2</sup> (95% CI  $0.83$ – $1.34$ ), whereas the supplementary analysis using the MMRM showed an ETD of  $1.07$  mL/min per  $1.73$  m<sup>2</sup> ( $0.82$ – $1.33$ ). The UACR from baseline to week 104 (FLOW and SELECT only) favoured semaglutide ( $0.92$  mg/g) over placebo ( $1.08$  mg/g), with an ETR of  $0.85$  (95% CI  $0.83$ – $0.88$ ); the corresponding supplementary analysis showed an ETR of  $0.85$  ( $0.83$ – $0.88$ ).

For the primary kidney composite outcome, baseline eGFR ( $<60$  mL/min per  $1.73$  m<sup>2</sup> vs  $\geq 60$  mL/min per  $1.73$  m<sup>2</sup>) did not interact with the treatment effect of semaglutide ( $p_{\text{interaction}}=0.168$ ). In participants with baseline eGFR less than  $60$  mL/min per  $1.73$  m<sup>2</sup> ( $n=7527$ ), 534 first events were observed in the semaglutide treatment group compared with 650 first events for placebo (IR  $4.1$  and  $5.2$  per 100 years of observation, respectively; HR  $0.79$  [95% CI  $0.70$ – $0.88$ ]). In participants with baseline eGFR of at least  $60$  mL/min per  $1.73$  m<sup>2</sup> ( $n=23038$ ), there were 432 versus 482 first events (IR  $1.1$  vs  $1.2$ ; HR  $0.89$

[ $0.78$ – $1.01$ ]; figure 2). Furthermore, no indication of treatment heterogeneity was observed for any secondary kidney outcomes according to baseline eGFR ( $<60$  mL/min per  $1.73$  m<sup>2</sup> or  $\geq 60$  mL/min per  $1.73$  m<sup>2</sup>;  $p_{\text{interaction}} > 0.1$ ; figure 2; appendix 1 p 19).

Consistent with the overall analysis, the ETD between semaglutide and placebo for the annual decline in eGFR was greater in participants with eGFR less than  $60$  mL/min per  $1.73$  m<sup>2</sup> ( $0.80$  mL/min per  $1.73$  m<sup>2</sup> per year [95% CI  $0.64$ – $0.95$ ]) versus those with eGFR of at least  $60$  mL/min per  $1.73$  m<sup>2</sup> at baseline ( $0.38$  mL/min per  $1.73$  m<sup>2</sup> per year [ $0.29$ – $0.47$ ];  $p_{\text{interaction}} < 0.0001$ ). Similarly, participants not using an SGLT2 inhibitor at baseline showed a greater ETD for annual decline in eGFR than those with SGLT2 inhibitor use (ETD of  $0.51$  mL/min per  $1.73$  m<sup>2</sup> per year [ $0.42$ – $0.59$ ] and  $0.25$  mL/min per  $1.73$  m<sup>2</sup> per year [ $0.02$ – $0.48$ ]; respectively;  $p_{\text{interaction}} = 0.04$ ; appendix 1 p 20). However, when annual rate of change in eGFR was assessed by baseline eGFR, UACR, or KDIGO risk category, no indication of heterogeneity was observed ( $p_{\text{interaction}} \geq 0.10$ ; appendix 1 p 20).

For change in eGFR from baseline to week 104, the ETD was greater in participants with eGFR less than  $60$  mL/min per  $1.73$  m<sup>2</sup> ( $2.41$  mL/min per  $1.73$  m<sup>2</sup>

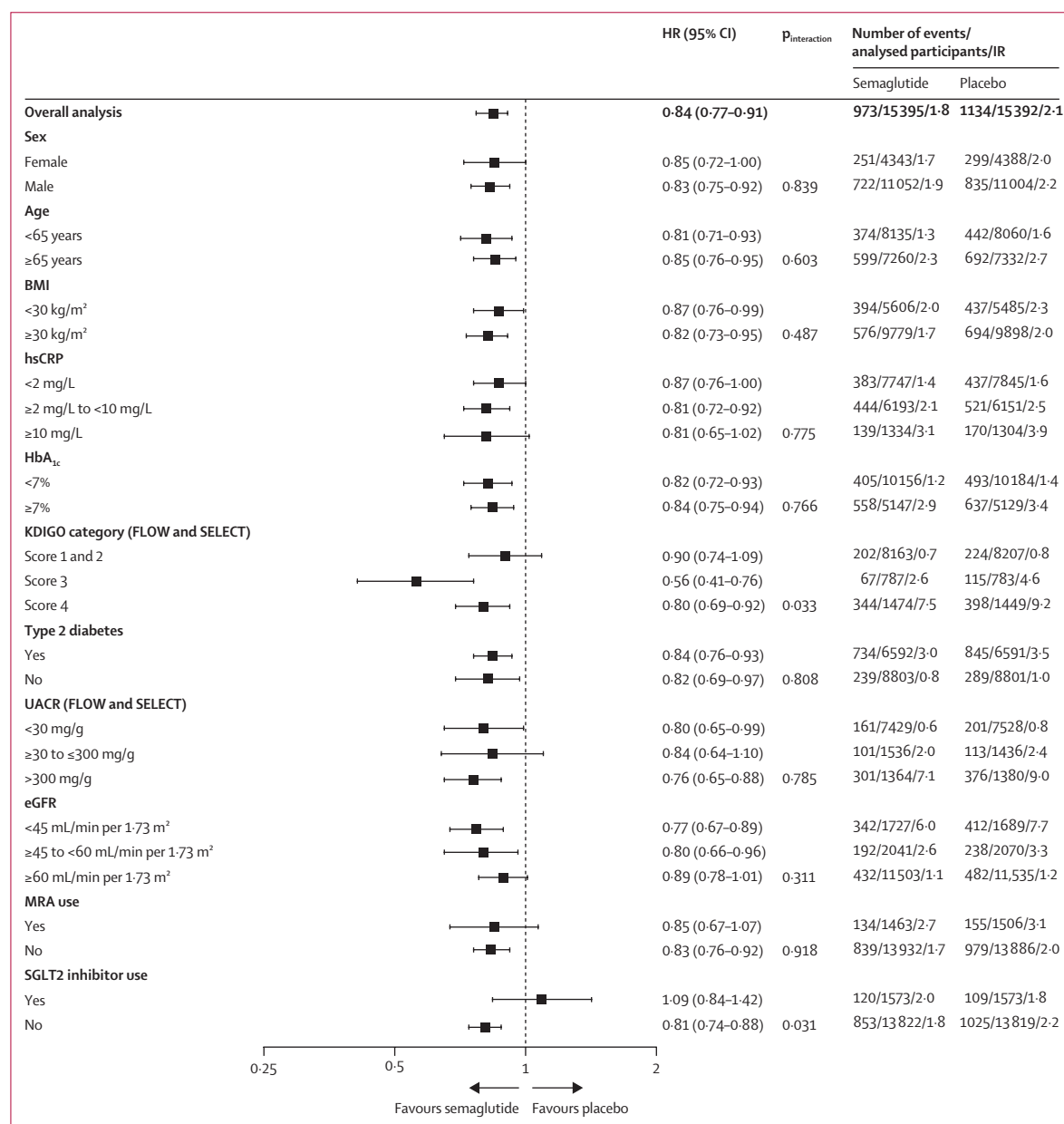


**Figure 2: Primary and secondary outcomes, overall and according to baseline eGFR in the pooled full analysis set**

Data are from the in-trial period. No kidney-related deaths were recorded for participants with eGFR  $\geq 60$  mL/min per  $1.73$  m<sup>2</sup> in the pooled semaglutide group, and in the pooled placebo group, three kidney-related deaths were recorded. Hence, the analysis of kidney-related death according to baseline eGFR was not feasible. Events were not included for participants with missing eGFR values in the subgroup analyses. In the FLOW trial, the analyses were stratified by use of SGLT2 inhibitors at randomisation. eGFR=estimated glomerular filtration rate. HR=hazard ratio. IR=incidence rate (number of events by 100 years of observation).

[1.89–2.93]) versus those with eGFR of at least 60 mL/min per 1.73 m<sup>2</sup> at baseline (0.67 mL/min per 1.73 m<sup>2</sup> [0.38–0.96];  $p_{\text{interaction}} < 0.0001$ ), with no indication of heterogeneity when analysed according to SGLT2 inhibitor use and CKD severity ( $p_{\text{interaction}} > 0.10$ ; appendix 1 pp 21–22). Supplementary MMRM analyses produced results consistent with the analysis of variance model using multiple imputation (appendix 1 pp 21–22).

For various participant subgroups, the effect of semaglutide on the primary kidney outcome was consistent with the overall analysis and was not affected by the sex, age, or BMI of participants, their baseline HbA<sub>1c</sub> or hsCRP concentrations, the presence of type 2 diabetes, or baseline use of MRAs. No indication of heterogeneity was observed for these subgroups ( $p_{\text{interaction}} > 0.10$ ; figure 3), except for KDIGO risk score and SGLT2 inhibitor use at baseline ( $p_{\text{interaction}}$  of 0.033 and 0.031, respectively).



**Figure 3: Time to first occurrence of the primary kidney composite in participant subgroups in the pooled full analysis set**

Data are from the in-trial period. The primary kidney composite included onset of persistent ≥50% reduction in eGFR, kidney failure (persistent eGFR <15 mL/min per 1.73 m<sup>2</sup> or initiation of kidney replacement therapy), kidney-related death, or cardiovascular-related death. KDIGO risk categories defined as low (1), moderate (2), high (3), and very high (4). All participants in the SELECT trial reported no baseline use of SGLT2 inhibitors; therefore, participants in SELECT were classified as not using SGLT2 inhibitors in the analyses. eGFR=estimated glomerular filtration rate. HR=hazard ratio. hsCRP=high-sensitivity C-reactive protein. IR=incidence rate (per 100 years of observation). KDIGO=Kidney Disease Improving Global Outcomes. MRA=mineralocorticoid receptor antagonist. UACR=urinary albumin-to-creatinine ratio.

Similar trends were observed for the narrower secondary kidney composite outcome in participant subgroups (appendix 1 pp 23–24).

Results from the fixed and random effects meta-analysis of the kidney composite outcomes were similar to those of the pooled analysis, with slightly wider CIs in the random effects meta-analysis (appendix 1 pp 25–26). No evidence of heterogeneity was observed among the three trials, including within the eGFR subgroups (<60 mL/min per 1.73 m<sup>2</sup> vs ≥60 mL/min per 1.73 m<sup>2</sup>; appendix 1 pp 27–28). In FLOW and SOUL, prespecified sensitivity analyses addressing dropout-related censoring and missing eGFR data were consistent with the primary analyses (appendix 1 p 6).

A total of 6130 (39.8%) of 15 395 participants assigned semaglutide reported SAEs compared with 6581 (42.8%) 15 392 participants assigned placebo (table 3). The most common (ie, reported in ≥5% of participants in either treatment group in the full analysis set) SAEs by system organ class were cardiac disorders, followed by infections and infestations, nervous system disorders, and neoplasms in both treatment groups. 2443 (15.9%) of 15 395 participants assigned semaglutide permanently discontinued treatment due to adverse events, compared with 1488 (9.7%) 15 392 participants assigned placebo. The most common adverse events leading to treatment discontinuation were classified as gastrointestinal disorders, reported in 1269 (8.2%) and 290 (1.9%) of participants in semaglutide and placebo groups, respectively. Acute renal failure (predefined MedDRA search) was reported for 522 (3.4%) of 15 395 participants in the semaglutide treatment group and 590 (3.8%) of 15 392 participants in the placebo group. For both semaglutide and placebo treatment groups, rates of SAEs were higher in participants with baseline eGFR less than 60 mL/min per 1.73 m<sup>2</sup> versus those with at least 60 mL/min per 1.73 m<sup>2</sup> (appendix 1 pp 7–9).

## Discussion

This prespecified analysis of participant-level data from over 30 000 participants in the SELECT, FLOW, and SOUL trials was undertaken to examine the long-term effects of semaglutide on clinically important kidney outcomes. The results from combining these large datasets with a mean follow-up of more than 3 years indicated that the kidney benefits of semaglutide appeared consistent regardless of kidney function, adiposity status, and type 2 diabetes history. In addition, the effect of semaglutide on kidney outcomes was consistent across the trials, despite different doses and routes of administration. The frequency of SAEs was numerically lower in the semaglutide group than in the placebo group, but more gastrointestinal adverse events leading to treatment discontinuation were reported for semaglutide in the full analysis set and across baseline

|                                                                                 | Semaglutide<br>(n=15 395) | Placebo<br>(n=15 392) |
|---------------------------------------------------------------------------------|---------------------------|-----------------------|
| SAEs                                                                            | 6130 (39.8%)              | 6581 (42.8%)          |
| Severity of SAEs                                                                |                           |                       |
| Mild                                                                            | 1276 (8.3%)               | 1280 (8.3%)           |
| Moderate                                                                        | 3403 (22.1%)              | 3672 (23.9%)          |
| Severe                                                                          | 3429 (22.3%)              | 3774 (24.5%)          |
| Most common* SAEs, by MedDRA SOC                                                |                           |                       |
| Cardiac disorders                                                               | 2142 (13.9%)              | 2457 (16.0%)          |
| Infections and infestations                                                     | 1665 (10.8%)              | 1908 (12.4%)          |
| Nervous system disorders                                                        | 977 (6.3%)                | 1027 (6.7%)           |
| Neoplasms benign, malignant, and unspecified                                    | 850 (5.5%)                | 773 (5.0%)            |
| Adverse events leading to treatment discontinuation                             | 2443 (15.9%)              | 1488 (9.7%)           |
| Most common* adverse events leading to treatment discontinuation, by MedDRA SOC |                           |                       |
| Gastrointestinal disorders                                                      | 1269 (8.2%)               | 290 (1.9%)            |
| Acute renal failure (predefined MedDRA search)                                  | 522 (3.4%)                | 590 (3.8%)            |
| Acute kidney injury                                                             | 402 (2.6%)                | 453 (2.9%)            |
| Renal impairment                                                                | 74 (0.5%)                 | 83 (0.5%)             |
| Renal failure                                                                   | 37 (0.2%)                 | 51 (0.3%)             |
| Adverse event of special interest                                               |                           |                       |
| COVID-19-related events                                                         | 605 (3.9%)                | 749 (4.9%)            |
| Malignant neoplasms                                                             | 749 (4.9%)                | 686 (4.5%)            |
| Gallbladder disorders                                                           | 243 (1.6%)                | 215 (1.4%)            |
| Acute pancreatitis                                                              | 57 (0.4%)                 | 57 (0.4%)             |
| Diabetic retinopathy†                                                           | 87 (0.6%)                 | 56 (0.4%)             |
| Cardiovascular disorder (SAE)‡                                                  | 272 (1.8%)                | 315 (2.0%)            |
| Gastrointestinal disorder (SAE)                                                 | 676 (4.4%)                | 627 (4.1%)            |
| Rare event (SAE)                                                                | 295 (1.9%)                | 306 (2.0%)            |
| Hepatic disorder (SAE)                                                          | 97 (0.6%)                 | 96 (0.6%)             |
| Allergic reaction (SAE)                                                         | 49 (0.3%)                 | 51 (0.3%)             |
| Abuse and misuse (SAE)                                                          | 13 (0.1%)                 | 13 (0.1%)             |
| Heart failure§                                                                  | 550 (3.6%)                | 644 (4.2%)            |
| Severe hypoglycaemia¶                                                           | 3 (<0.1%)                 | 1 (<0.1%)             |
| Deaths                                                                          | 1130 (7.3%)               | 1314 (8.5%)           |
| Kidney-related deaths                                                           | 6 (<0.1%)                 | 12 (0.1%)             |

Data are n (%) and from the in-trial period. All permanent treatment discontinuations until the end of treatment visit were included. For deaths and kidney deaths, only one event per participant was counted. Kidney deaths were adjudicated. Adverse events of special interest categories were derived using predefined MedDRA search flags defined within each individual trial.

eGFR=estimated glomerular filtration rate. MedDRA=Medical Dictionary for Regulatory Activities (version 26.0 [SELECT]; 27.0 [SOUL], and 26.1 [FLOW]). SAE=serious adverse event. SOC=system organ class. \*Reported in ≥5% of participants in either treatment group in the full analysis set. †Diabetic retinopathy was not defined for SELECT. ‡Cardiovascular disorder (SAE) was not defined for SELECT and SOUL. §High-level group term contained heart failure. ¶Severe hypoglycaemia was not defined for SOUL and FLOW.

**Table 3: Safety outcomes with pooled data from the SELECT, FLOW, and SOUL trials in the full analysis set**

eGFR subgroups. Acute renal failure was reported with similar frequency in both treatment groups overall and in subgroups with eGFR less than 60 mL/min per 1.73 m<sup>2</sup>.

Semaglutide reduced the risk of kidney outcomes compared with placebo on the basis of the primary standard composite outcome used for kidney outcome trials, and on a narrower secondary kidney-specific composite of CKD that excluded cardiovascular-related death. Therefore, although cardiovascular-related death contributed substantially to the number of primary kidney outcome events, the benefit of semaglutide was also observed for the more specific kidney endpoints and for the individual components of the primary composite outcome. Similar results were observed when the kidney composites were defined according to differing thresholds of eGFR reductions ( $\geq 30\%$ ,  $\geq 40\%$ , and  $\geq 57\%$ ). The decline in eGFR, measured as annual rate of change in eGFR, was slower with semaglutide than with placebo, and the benefit of semaglutide on the progression of kidney disease appeared greater in the subgroup with eGFR less than 60 mL/min per 1.73 m<sup>2</sup>.

In addition to semaglutide, data from cardiovascular outcomes trials conducted with other GLP-1 receptor agonists suggest that these drugs could provide kidney-protective effects.<sup>20</sup> Such data are mainly based on a reduction in albuminuria as a surrogate marker, a less robust outcome. Subgroup analyses of trials with dulaglutide and liraglutide have suggested an attenuation of eGFR decline and reduction in other kidney outcomes, which support the findings of the dedicated kidney outcomes trial with semaglutide (FLOW).<sup>12,21,22</sup> Tirzepatide, a GLP-1 receptor agonist and a glucose-dependent insulinotropic polypeptide receptor agonist, might prevent eGFR loss, at least short term, in people with obesity, type 2 diabetes, or both, but we have found no dedicated long-term kidney outcome trials that have been reported.<sup>23</sup> The SURPASS cardiovascular outcomes trial included kidney outcomes as hypothesis-generating secondary endpoints only.<sup>24</sup>

The large number of participants in our analyses allowed for the exploratory examination of subgroups of clinical interest. The relative benefits of semaglutide on kidney outcomes were independent of a baseline eGFR above or below 60 mL/min per 1.73 m<sup>2</sup> (figure 2; appendix 1 p 19), but, as previously indicated, the absolute benefit of semaglutide on eGFR decline could be somewhat greater in people with a baseline eGFR less than 60 mL/min per 1.73 m<sup>2</sup>. Other baseline characteristics, including UACR, HbA<sub>1c</sub>, BMI, age, or sex had no effect on the relative benefit of semaglutide over placebo for the kidney composite outcomes (figure 3; appendix 1 p 23). Two subgroups emerged with a significant  $p_{\text{interaction}}$ . First, the analyses indicated that the effect of semaglutide was more pronounced in participants classified as high risk according to baseline KDIGO risk category, and a consistent but smaller effect was observed in those with very high and low or moderate risk classification. Second, the benefit of semaglutide as measured with the primary kidney composite outcome could not be shown with confidence in participants using

SGLT2 inhibitors at baseline. However, semaglutide blunted the eGFR loss versus placebo in those treated with SGLT2 inhibitors and lowered albuminuria. Therefore, the totality of evidence supports the combined use of semaglutide with SGLT2 inhibitors for kidney protection. Notably, the benefits of semaglutide on cardiovascular and mortality outcomes showed no interaction with SGLT2 inhibitor use at baseline in the individual studies.<sup>25,26</sup> Combining GLP-1 receptor agonists and SGLT2 inhibitors is supported by meta-analyses of cardiovascular outcomes trials with SGLT2 inhibitors<sup>27</sup> and with GLP-1 receptor agonists,<sup>28</sup> which showed no effect modification in subgroups using the alternative agent. These meta-analyses complement a large Clinical Practice Research Datalink analysis from the UK in which the combination of GLP-1 receptor agonists and SGLT2 inhibitors was associated with a lower risk of serious kidney outcomes compared with either drug class alone.<sup>29</sup>

The size of the combined dataset is a major strength of this pooled analysis, as are the similarities shared by the SELECT, FLOW, and SOUL trials in terms of their design, objectives, and methodologies, despite notable differences in participant characteristics. Therefore, and unlike previous study-level meta-analyses, we were able to apply uniform definitions of kidney outcomes to the combined dataset. Combining participant-level data allowed for subgroup analyses across a range of baseline parameters. It is important to note, however, that SOUL did not measure UACR, which limits the analysis of UACR as a determinant of kidney risk. In this pooled analysis, eGFR was calculated based on serum creatinine measurements. Caution is advised when interpreting such data in the context of weight loss, given the potential for lower creatinine production due to lean muscle loss.<sup>30</sup> Furthermore, eGFR is indexed to body surface area, which decreases with reductions in bodyweight. Of note, the primary analysis of FLOW, in which eGFR was also calculated on the basis of cystatin C and creatinine concentrations, showed findings consistent with both measurements.<sup>12</sup> A mediation analysis of FLOW data indicated that bodyweight change did not mediate eGFR change.<sup>31</sup> However, in SELECT, a similar analysis indicated that bodyweight change explained 81% of the eGFR change on the basis of creatinine measurements, but the correlation coefficient between individual changes in bodyweight and eGFR was only  $-0.053$ .<sup>18</sup> To add complexity, in SELECT, eGFR increased when baseline eGFR was low ( $<60$  mL/min per 1.73 m<sup>2</sup>) and decreased when baseline eGFR was high ( $\geq 60$  mL/min per 1.73 m<sup>2</sup>), whereas bodyweight decreased consistently by eGFR group.<sup>18,32</sup>

A further caveat when interpreting the overall effects of semaglutide is the considerable heterogeneity of the trial populations. That heterogeneity and—as in most recent kidney outcome trials—the inclusion of cardiovascular-related death in the primary kidney outcome,

must be considered when making clinical inferences. For example, cardiovascular-related death contributed 37% and 74% of the primary outcomes in FLOW and SOUL, respectively. It is reassuring that the narrower, kidney-specific outcomes are in parallel with the primary kidney outcome, although with fewer events and hence less precision in the estimates. Kidney failure had a modest event number, occurring in 206 (1.3%) of 15 395 participants receiving semaglutide versus 241 (1.6%) of 15 392 participants receiving placebo. We did analyses in various subgroups and with various outcomes. As in other combined analyses, we did not correct for multiple testing. Therefore, results should be understood as hypothesis-generating rather than definitive.

Missing data are unlikely to have materially influenced the estimated effect on the primary kidney outcome. Across trials, overall study completion was high ( $\geq 96.9\%$ ) and vital status was known for almost all participants ( $>97\%$ ), with low and broadly balanced rates of censoring before end of study between treatment groups.<sup>12–14</sup> In addition, prespecified sensitivity analyses addressing both dropout-related censoring and missing eGFR values gave results that were essentially unchanged from the primary analyses in the trials in which these analyses were performed. Taken together, these findings suggest that any bias introduced by missingness was likely small.

In conclusion, the data suggest that semaglutide prevents composite kidney outcomes and blunts decline in eGFR, if we combine data from three major long-term outcome trials conducted in populations with varying kidney function, adiposity, and diabetes status, and across oral and subcutaneous formulations. The kidney benefits are particularly apparent in high-risk populations with diabetes and CKD and hence substantial incidence rates of outcomes. Risk estimates are less precise in those participants without CKD and without type 2 diabetes or adiposity, but benefits tend to the same direction as in participants with CKD. These findings are consistent with current guidelines of kidney, cardiovascular, and diabetes organisations that recommend use of GLP-1 receptor agonists, nonsteroidal MRAs, and SGLT2 inhibitors in people with CKD.<sup>33</sup>

#### Contributors

This pooled analysis was designed by TI and one Novo Nordisk employee (not included as an author on this publication). JFEM and SR collected, assembled, and analysed the data for this pooled analysis. All authors had access to and interpreted the data, reviewed and revised the manuscript, provided final approval of the manuscript, and had responsibility for the decision to submit for publication.

#### Declaration of interests

JFEM declares grants from Novo Nordisk; consulting fees from Bayer and Novo Nordisk; honoraria from AstraZeneca, Bayer, Cytel, Hexal, ICON, IQVIA, Novo Nordisk, UpToDate, and WCG; has participated on a data safety monitoring board or advisory board for AstraZeneca, Bayer, and Sanofi; and has a leadership role in the KDIGO group. SVB declares consulting fees from AstraZeneca, Bayer, GSK, and Vifor Pharma; speaking fees from AstraZeneca, Bayer, Pfizer, and Vifor Pharma

(all honoraria paid to institution); and non-financial research support from Bayer. KB-F, TI, and GKH are former employees of Novo Nordisk. JBB declares salary support from clinical trial contracts to his employer from Corcept Therapeutics, Dexcom, GentiBio, and Novo Nordisk; is a consultant with personal compensation from Aardvark Therapeutics, Altimune, Alveus Therapeutics, Amgen, Antag Therapeutics, Aqua Medical, AstraZeneca, Boehringer Ingelheim, Corcept Therapeutics, Dexcom, Eli Lilly, embecta, General Medicines, GentiBio, Insulet, Kayothera, Metsera, MiniMed, Novo Nordisk, Recordati, Sparrow Pharmaceuticals, Tandem, Vertex, vTv Therapeutics, and Zealand Pharma; and holds stock options in Glyscend, Mellitus Health, Metsera, Pendulum Therapeutics, Praetego, and Stability Health. HMC declares consulting fees from Roche and Sanofi; serving on advisory panels for Bayer and Novo Nordisk; research grants from Chief Scientist Office, Diabetes UK, European Commission, IQVIA, Breakthrough T1D, Medical Research Council, and Sanofi; support to attend meetings/travel from Novo Nordisk; and payment or honoraria from Medscape. JED declares consulting fees from Aegerion, Amgen, AstraZeneca, Bayer, Boehringer Ingelheim, Merck, Novartis, Novo Nordisk, Pfizer, Sanofi, and Takeda; payment or honoraria from Aegerion, Amgen, AstraZeneca, Bayer, Boehringer Ingelheim, Merck, Novartis, Novo Nordisk, Pfizer, Sanofi, and Takeda; and a research grant from Alzheimer's Research UK. AML declares consulting honoraria from Akebia, Alnylam, Amgen, Ardelyx, Becton Dickinson, BrainStorm Cell Therapeutics, Canary Cure, Eli Lilly, Endologix, FibroGen, GSK, Intarcia, Medtronic, Neovasc, Novo Nordisk, Provention Bio, and Recor; and research funding (to institution) from AbbVie, AstraZeneca, CSL Behring, Eli Lilly, Esperion, and Novartis. IL declares research funding (paid to institution) and trial product related to investigator initiated studies from Novo Nordisk; clinical trial work for Boehringer Ingelheim, Dexcom, Eli Lilly, Novo Nordisk, Pfizer, and Roche; research-related consulting fees (paid to institution) from Novo Nordisk; and travel and publication support from AbbVie, Altimune, Alveus Therapeutics, Amgen, Antag Therapeutics, AstraZeneca, Bain Capital, Bayer, Betagenon, BIOIO, Biomea Fusion, Boehringer Ingelheim, Carmot Therapeutics, Corcel, Cytokine Pharma, Eli Lilly, Genentech, Intercept, Janssen/Johnson & Johnson, Juvena, Keros Therapeutics, Mediflix, Merck, Metsera, Neurocrine, Novo Nordisk, Pfizer, Regeneron, Roche, Sanofi, Shionogi, Source Bioscience, Structure Therapeutics, TERNs Pharma, The Comm Group, Verdiva Bio, WebMD, and Zealand Pharma. KWM declares grants from the American College of Cardiology, American Heart Association, Apple, California Institute for Regenerative Medicine, CSL Behring, Cytokinetics, Eidos, Element, Google (Verily), Idorsia, Johnson & Johnson, Novartis, PAC-12, Precordior, and St Jude Medical; consulting fees from Applied Therapeutics, BridgeBio, Bristol Myers Squibb, Eli Lilly, Elsevier, Fosun Pharma, Human, Moderna, MyoKardia, Novo Nordisk, PhaseBio, Quidel, Scleroderma Foundation, and Theravance; payment or honoraria from CSL Behring; and equity in Human, Medeloo, Precordior, and Regencor. DKM declares consulting fees from Alveus Therapeutics, Amgen, Boehringer Ingelheim, F Hoffmann La Roche, Eli Lilly, Kailera, Lexicon, Metsera, Neurotronic, Novo Nordisk, Regor, and Zealand Pharma; trial leadership for Amgen, Boehringer Ingelheim, Eli Lilly, Metsera, NewAmsterdam Pharma, Novartis, Novo Nordisk, and Pfizer; and research support from Amgen, Boehringer Ingelheim, Eli Lilly, Metsera, NewAmsterdam Pharma, Novartis, Novo Nordisk, and Pfizer. REP declares speaker fees from Corcept Therapeutics, Lilly USA, and Novo Nordisk; consulting fees from AbbVie, Altanine, AstraZeneca, Bayer, Bayer HealthCare Pharmaceuticals, Corcept Therapeutics, Dexcom, Endogenex, Gashierbrum Bio, Genprex, Getz Pharma, Hanmi Pharmaceutical, Hengrui, Intas Pharmaceuticals, Lilly USA, Merck, Novo Nordisk, Pfizer, Rivus Pharmaceuticals, Scholar Rock, and Sun Pharmaceutical Industries; and investigator-initiated grant support from Carmot Therapeutics, Dompé, Endogenex, Fractyl, Eli Lilly, Novo Nordisk, Roche, and Sanofi. PR declares consultancy and/or speaking fees (paid to institution) from Abbott, AstraZeneca, Bayer, Boehringer Ingelheim, Eli Lilly, Gilead, Novo Nordisk, and Sanofi; research grants from AstraZeneca, Bayer, and Novo Nordisk; and study drugs for investigator-initiated studies from Bayer, Novo Nordisk, and Lexicon Pharmaceuticals. NS declares consulting fees and/or speaker honoraria from AbbVie, Amgen, AstraZeneca, Boehringer Ingelheim, Carmot Therapeutics, Eli Lilly, Gan & Lee, GSK, Hanmi Pharmaceutical, Kailera,

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#### Data sharing

An authorised researcher can request access to clinical study data by submitting a research proposal for review and approval by Novo Nordisk and an internal independent review panel. Requests are usually considered after the research is finished and the main results have been published. If the research supports a regulatory application, requests will be considered after the product and its intended use are approved in both the EU and the USA. Participants' clinical data will be anonymised, following an approved internal process, before data are shared to external third parties. For details on how to request access to clinical data, visit <https://www.novonordisk-trials.com>.

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